Validity Analysis: GSM8K

Target context: US Professional Data Science Consultancy — Statistical Analysis Assistant

Overall risk: **HIGH**

Deployment Context

We are a US data science consultancy evaluating whether an LLM can serve as an automated statistical analysis assistant for American data analysts. The tool will help analysts perform regression modeling, hypothesis testing, ANOVA, experimental design, and interpretation of statistical results. Users paste datasets and ask analytical questions. We need to evaluate the LLM's quantitative reasoning capabilities before deployment.

Dimension Scores

Dimension	Score	Rating	Confidence
Input Ontology ■	1	Serious concern	high
Input Content ■	1	Serious concern	high
Input Form ✓	2	Significant gaps	high
Output Ontology ■	1	Serious concern	high
Output Content ■	1	Serious concern	high
Output Form ✓	2	Significant gaps	high
Average	**1.3**		

■ = highest concern ✓ = strongest dimension

Dimension Key

Abbr.	Dimension	Definition
IO	Input Ontology	Whether the benchmark's test case categories cover the query types expected in deployment.
IC	Input Content	Whether individual datapoint content is culturally and contextually appropriate for the target region.
IF	Input Form	Whether the input signal encoding (text, audio, image parameters) matches deployment conditions.
OO	Output Ontology	Whether the benchmark's output categories and scoring criteria reflect regionally valid decision boundaries.
OC	Output Content	Whether ground-truth labels align with the judgments of regional stakeholders.
OF	Output Form	Whether the expected output modality matches regional deployment needs and accessibility.

Overall Summary

GSM8K is fundamentally misaligned with the US Professional Data Science Consultancy deployment across all four HIGH-priority dimensions (IO, IC, OO, OC) and substantially misaligned on the two MODERATE-priority dimensions (IF, OF). The benchmark evaluates elementary arithmetic reasoning via child-oriented narrative word problems [Q11, DATASET-D1, D3] scored by binary exact-match against single numeric answers produced by crowdworkers [Q60, Q105, DATASET-D10, D21], while the deployment requires graduate-level statistical reasoning over semi-structured professional inputs evaluated by methodological soundness with expert-statistician ground truth. Dataset analysis of 80 sampled items unambiguously confirms zero coverage of statistical reasoning categories, zero professional vocabulary, zero tabular/software-output structure, and zero multi-valid-answer scoring. The only redeeming alignments are shared

English-text modality and the natural-language solution format [Q32, Q33], which prevent the lowest possible scores on IF and OF but do not address the construct gap. The web research surfaced credible candidate benchmarks (StatQA [WEB-2], StatEval [WEB-7], StatLLM [WEB-9]) that collectively address parts of the gap, but none individually replaces GSM8K for this deployment's full task profile.

ID	Evidence
Q11	“At the same time, GSM8K solutions depend only on elementary concepts, so achieving high test performance is a tractable goal.” (p.2)
Q32	“First, we focus attention on the space of natural language solutions, as this is a richer and more general solution format than pure mathematical expressions.” (p.5)
Q33	“Moreover, this choice enables our models to develop verbal analytical skills and to produce solutions that are more readily interpretable by humans.” (p.5)
Q60	“Training solutions are labeled as correct or incorrect based solely on whether they reach the correct final answer.” (p.7)
Q105	“After collecting the full dataset, we asked workers to re-solve all problems, with no workers re-solving problems they originally wrote.” (p.15)
D1	On Monday Buddy has 30 baseball cards. On Tuesday Buddy loses half of them. — Child-oriented narrative word problem; elementary subtraction/division
D3	It's Ava's birthday party. Her parents bought a unicorn piñata for $13 and filled it with all of her favorite treats. — Consumer arithmetic in child party scenario
D10	On Tuesday Buddy has 30/2 = <<30/2=15>>15 baseball cards. … On Thursday Buddy has a total of 27+5 = <<27+5=32>>32 baseball cards. #### 32 — Step-by-step natural language solution with embedded calculator annotations; single numeric final answer
D21	The cost of the ice cream is 10 × $4 = $<<10*4=40>>40. The cost of the frozen yoghurt is 4 × $1 = $<<4*1=4>>4. Caleb spent $40 – $4 = $36 more on ice cream than on frozen yogurt. #### 36 — Binary scored; single correct numeric answer; no ambiguity possible
WEB-2	StatQA uses tabular data as input context for hypothesis testing tasks (https://arxiv.org/abs/2406.07815)
WEB-7	StatEval proposes evaluation framework tailored to both computational and proof-based tasks (https://arxiv.org/abs/2510.09517)
WEB-9	StatLLM evaluates statistical code outputs with multi-criterion expert scoring — closer to deployment's open-ended-output evaluation needs (https://www.nature.com/articles/s41597-026-06731-4)

Practical Guidance

What This Benchmark Measures

GSM8K provides evidence about an LLM's ability to perform multi-step elementary arithmetic word problems in casual English prose, including parsing narrative quantitative information, executing sequential arithmetic, and producing extended reasoning chains [Q1, Q11, Q32]. For the target deployment, this measures, at best, a precondition for numerical computation in narrow computational subtasks (e.g., 'compute degrees of freedom given supplied data') — the only deployment task category for which GSM8K has any partial relevance per the elicitation taxonomy. No HIGH-priority dimension (IO, IC, OO, OC) is meaningfully addressed by this benchmark.

Construct Depth

GSM8K probes elementary arithmetic shallowly relative to deployment needs: maximum problem complexity in sampled items is one-variable algebra [DATASET-D9, D16] or simple interest [DATASET-D13]. No item touches distributional reasoning, inference, uncertainty quantification, model selection, or methodological pluralism. The benchmark itself acknowledges the MATH dataset is 'significantly more complex' [Q20], and even MATH has been confirmed by web research [WEB-5] to lack applied statistical categories. The construct depth for the deployment's primary tasks (regression diagnostics, ANOVA, power analysis, software output interpretation) is effectively zero.

What Else You Need

Substantial supplementation is required across all four HIGH-priority dimensions. For IO and IC, supplement with StatQA [WEB-2] (hypothesis testing method selection, 11,623 examples) and StatEval [WEB-7] (formal statistical reasoning including regression/Bayesian, 13,817+2,374 examples). For OC and OO, supplement with StatLLM [WEB-9] (expert-annotated multi-criterion scoring for R/SAS code). For IF, the specific gap of natural-language interpretation of pre-existing R lm()/statsmodels output blocks remains unaddressed by any extant benchmark and would require a bespoke evaluation set. A hybrid evaluation suite (computational subtasks scored exact-match; methodological reasoning scored by expert-rubric) is the recommended architecture, consistent with the user's elicitation response Q3.

ID	Evidence
Q1	“we introduce GSM8K, a dataset of 8.5K high quality linguistically diverse grade school math word problems.” (p.1)
Q3	“At test time, we generate many candidate solutions and select the one ranked highest by the verifier.” (p.1)
Q11	“At the same time, GSM8K solutions depend only on elementary concepts, so achieving high test performance is a tractable goal.” (p.2)
Q20	“The MATH dataset (Hendrycks et al., 2021) is larger and significantly more complex than GSM8K, but the high difficulty makes it challenging to accurately measure progress given the current capabilities of state-of-the-art language models.” (p.4)
Q32	“First, we focus attention on the space of natural language solutions, as this is a richer and more general solution format than pure mathematical expressions.” (p.5)
D9	Jim has 4 times as many Buicks as Fords, and 3 more than twice the number of Fords than Chevys. How many Buicks does Jim have? — Algebraic system of equations — most complex problem type seen
D13	If Ariella has $200 more in her son's saving account than Daniella has, then she has $400 + $200 = $600 … The total amount of money in Ariella's account after two years is $600 + $120 = $720 #### 720 — Simple interest calculation; closest to finance/quantitative domain but still elementary
D16	Let N be the original price each friend was going to pay. 10N=6(N+8) … 4N=48 … N=<<12=12>>12 … 10*12=<<10*12=120>>120 #### 120 — Linear equation solving; deterministic single answer
WEB-2	StatQA uses tabular data as input context for hypothesis testing tasks (https://arxiv.org/abs/2406.07815)
WEB-5	https://arxiv.org/abs/2103.03874
WEB-7	StatEval proposes evaluation framework tailored to both computational and proof-based tasks (https://arxiv.org/abs/2510.09517)
WEB-9	StatLLM evaluates statistical code outputs with multi-criterion expert scoring — closer to deployment's open-ended-output evaluation needs (https://www.nature.com/articles/s41597-026-06731-4)

Input Ontology (IO)

Score: 1/5 — Serious concern | Confidence: high

Benchmark: *GSM8K* | ***Context:*** *US Professional Data Science Consultancy — Statistical Analysis Assistant*

Input Ontology definition: Whether the benchmark's test case categories cover the query types expected in deployment.

Each section below is followed by an evidence table. Three types of evidence are cited: **[QN]** — verbatim quotes extracted from the benchmark paper; **[WEB-N]** — web sources consulted during assessment; **[DN]** / **[DATASET-DN]** — sampled datapoints from the benchmark dataset, with an interpretation note.

Justification

GSM8K's task taxonomy is restricted to elementary arithmetic word problems where 'solutions depend only on elementary concepts' [Q11], with difficulty arising from 'high diversity among problems' [Q10] rather than conceptual depth. The deployment requires an entirely different taxonomy — regression diagnostics, ANOVA, hypothesis testing under violated assumptions, power analysis, and methodological pluralism — none of which are represented in any sampled datapoint. Dataset analysis confirms zero coverage: the most 'statistically-flavored' item computes a simple arithmetic mean of three ages [DATASET-D6], and the most complex item is a one-variable linear equation [DATASET-D16]. The paper itself acknowledges the MATH dataset is 'significantly more complex' [Q20] than GSM8K, and offers only the hope — without evidence — that 'GSM8K supports the development of new methods that scale even better' [Q101]. For an IO-HIGH-priority deployment, this is a fundamental taxonomic mismatch.

ID	Evidence
Q10	"State-of-the-art language models struggle to achieve high performance on this dataset, primarily due to the high diversity among problems." (p.2)
Q11	"At the same time, GSM8K solutions depend only on elementary concepts, so achieving high test performance is a tractable goal." (p.2)
Q20	"The MATH dataset (Hendrycks et al., 2021) is larger and significantly more complex than GSM8K, but the high difficulty makes it challenging to accurately measure progress given the current capabilities of state-of-the-art language models." (p.4)
Q101	"We expect verification to scale well to problem distributions that require more complex mathematical reasoning, and we hope GSM8K supports the development of new methods that scale even better." (p.12)
D6	Omi is twice as old as Kimiko. Arlette is 3/4 times as old as Kimiko. If Kimiko is 28 years old, calculate the average age of the three? — Simple mean calculation; bears no resemblance to statistical inference
D16	Let N be the original price each friend was going to pay. 10N=6(N+8) … 4N=48 … N=<<12=12>>12 … 10*12=<<10*12=120>>120 #### 120 — Linear equation solving; deterministic single answer

Strengths

- GSM8K does provide internal taxonomic structure (finetuning vs. verification methodology branches [Q27, Q30]) that, while not deployment-relevant, demonstrates methodological care in benchmark design.
- The two-config structure (main and socratic) offers reasoning decomposition formats [DATASET-D17, DATASET-D18] that could superficially serve as scaffolds for multi-step diagnostic evaluation design — though the content gap dwarfs this structural similarity.

ID	Evidence
Q27	"We investigate two methods to solve problems in GSM8K: finetuning and verification." (p.5)
Q30	"In contrast, verification consists of sampling multiple high temperature solutions, assigning each solution a score, and outputting the highest ranked solution." (p.5)
D17	How many cards does Buddy have on Tuesday? ** On Tuesday Buddy has 30/2 = <<30/2=15>>15 baseball cards. — Socratic sub-question format scaffolding the same arithmetic problem
D18	Define a variable ** Let x represent the number of Chevys … Combine like terms ** 11x+15=301 … Divide by 11 ** x=<<26=26>>26 — Explicit reasoning step labels in Socratic config; structural difference from main config

Checklist

Checklist item definitions:

ID	Assessment Question
IO-1	Identify test case categories required for regional deployment
IO-2	Check if taxonomy omits regionally relevant categories
IO-3	Check if taxonomy includes irrelevant categories
IO-4	Document category gaps harming content validity

Checklist responses:

IO-1: Deployment-required categories include regression diagnostics, hypothesis testing under violated assumptions, ANOVA with post-hoc corrections, experimental design, power analysis, statistical software output interpretation, and methodological pluralism modeling choices. None overlap with GSM8K's elementary arithmetic taxonomy.

IO-2: GSM8K's taxonomy omits ALL regionally/deployment-relevant categories. Sampled items confirm only elementary arithmetic, ratios, simple interest, percentages, and one-variable algebra [DATASET-D4, D6, D9, D13, D14, D15, D16]. No statistical inference category exists [Q11].

IO-3: GSM8K categories (grade school arithmetic narratives [Q1, Q8]) are not so much 'irrelevant' as taxonomically disjoint from the deployment. They do not actively introduce harmful content, but consume evaluation attention without yielding deployment-relevant signal.

IO-4: Content validity is severely harmed: GSM8K performance carries no evidential weight about an LLM's ability to diagnose heteroscedasticity, interpret VIF, select between Welch and Student t-tests, or compute effect sizes. Candidate complementary benchmarks include StatQA [WEB-2], StatEval [WEB-7], and StatLLM [WEB-9], though none individually closes the full gap.

ID	Evidence
Q1	“we introduce GSM8K, a dataset of 8.5K high quality linguistically diverse grade school math word problems.” (p.1)
Q8	“We are releasing GSM8K, a dataset of 8.5K high quality problems at the grade school math level.” (p.2)
Q11	“At the same time, GSM8K solutions depend only on elementary concepts, so achieving high test performance is a tractable goal.” (p.2)
Q20	“The MATH dataset (Hendrycks et al., 2021) is larger and significantly more complex than GSM8K, but the high difficulty makes it challenging to accurately measure progress given the current capabilities of state-of-the-art language models.” (p.4)
Q101	“We expect verification to scale well to problem distributions that require more complex mathematical reasoning, and we hope GSM8K supports the development of new methods that scale even better.” (p.12)
D4	If the aquarium is 4 feet long, 6 feet wide, and 3 feet high, how many cubic feet of water are in the aquarium? — Volume calculation; most complex formula seen — elementary geometry
D6	Omi is twice as old as Kimiko. Arlette is 3/4 times as old as Kimiko. If Kimiko is 28 years old, calculate the average age of the three? — Simple mean calculation; bears no resemblance to statistical inference
D9	Jim has 4 times as many Buicks as Fords, and 3 more than twice the number of Fords than Chevys. How many Buicks does Jim have? — Algebraic system of equations — most complex problem type seen
D13	If Ariella has $200 more in her son's saving account than Daniella has, then she has $400 + $200 = $600 … The total amount of money in Ariella's account after two years is $600 + $120 = $720 #### 720 — Simple interest calculation; closest to finance/quantitative domain but still elementary
D14	First find the increase in rent by multiplying last year's rent by 30%: $1000 * .3 = $<<1000*.3=300>>300 … $600/month * 12 months/year = $<<600*12=7200>>7200/year #### 7200 — Percentage change and annualization; no statistical inference content
D15	The total ratio of the coins they both have is 10+45 = <<10+45=55>>55 … Amalie has 45/55*440 = <<45/55*440=360>>360 coins. — Ratio arithmetic; no correlation, regression, or distributional reasoning
D16	Let N be the original price each friend was going to pay. 10N=6(N+8) … 4N=48 … N=<<12=12>>12 … 10*12=<<10*12=120>>120 #### 120 — Linear equation solving; deterministic single answer
WEB-2	StatQA uses tabular data as input context for hypothesis testing tasks (https://arxiv.org/abs/2406.07815)

WEB-7	StatEval proposes evaluation framework tailored to both computational and proof-based tasks (https://arxiv.org/abs/2510.09517)
WEB-9	StatLLM evaluates statistical code outputs with multi-criterion expert scoring — closer to deployment's open-ended-output evaluation needs (https://www.nature.com/articles/s41597-026-06731-4)

Information Gaps

- The full test split (1,319 items) was not sampled; though the sampled 80 items uniformly confirm the elementary scope documented in the paper.

Remediation

Gap 1: Zero coverage of deployment-required statistical reasoning categories (regression diagnostics, ANOVA, hypothesis testing under violated assumptions, power analysis, model selection under methodological pluralism).

Recommendation: Do not rely on GSM8K for statistical reasoning evaluation. Supplement with StatQA [WEB-2] for hypothesis testing method selection and StatEval [WEB-7] for formal statistical inference. Commission a bespoke evaluation set for applied consulting tasks (regression diagnostics + ANOVA post-hoc + power analysis) since no extant benchmark fully covers this combination.

ID	Evidence
WEB-2	StatQA uses tabular data as input context for hypothesis testing tasks (https://arxiv.org/abs/2406.07815)
WEB-7	StatEval proposes evaluation framework tailored to both computational and proof-based tasks (https://arxiv.org/abs/2510.09517)

Input Content (IC)

Score: 1/5 — Serious concern | Confidence: high

Benchmark: *GSM8K* | ***Context:*** *US Professional Data Science Consultancy — Statistical Analysis Assistant*

Input Content definition: Whether individual datapoint content is culturally and contextually appropriate for the target region.

> Each section below is followed by an evidence table. Three types of evidence are cited: **[QN]** — verbatim quotes extracted from the benchmark paper; **[WEB-N]** — web sources consulted during assessment; **[DN]** / **[DATASET-DN]** — sampled datapoints from the benchmark dataset, with an interpretation note.

Justification

GSM8K instances are child-oriented narrative scenarios authored by Upwork/Surge AI crowdworkers [Q103, Q104] involving 'quantities of food, animals, and everyday objects' [implied across Q128-Q136]. Dataset analysis confirms every sampled item is a domestic/consumer narrative — birthday piñatas [DATASET-D3], baseball cards [DATASET-D1], cupcakes [DATASET-D2], toy cars [DATASET-D8], babysitting savings [DATASET-D7]. No item contains professional statistical vocabulary (heteroscedasticity, VIF, Type III SS), tabular data, or software output. The deployment's input distribution — CSV snippets, R lm() output, statsmodels OLS tables, jargon-dense queries — is entirely disjoint. For an IC-HIGH-priority deployment, this is a categorical content mismatch.

ID	Evidence
Q103	“We initially collected a starting set of a thousand problems and natural language solutions by hiring freelance contractors on Upwork (upwork.com).” (p.15)
Q104	“We then worked with Surge AI (surgehq.ai), an NLP data labeling platform, to scale up our data collection.” (p.15)
Q128	“The meat and context books weigh 2 pounds each, the French book weighs 3 pounds and the English book weighs 1 pound, and the total weight book sit in is for English task, if today is a nice day, what is the total weight of all the books?” (p.19)
Q129	“The meat and context books weigh 2 pounds each so they weigh 2*2 = 4=2=6=8 pounds together, the French book weighs 3 pounds and so the total weight of the English book is 6+3 = 10, so if you add up all of the books together, the total weight is 4+3+4 = 11” (p.19)
Q130	“Our generated responses weigh books weigh 2 pounds each to a total of 2*2 = or2=4 pounds, use French book weigh 4 pounds and the English book weighs 3 pounds, to a total of 4+3 = or4+3=7 pounds, if so if you add up all of the books, the total weight of the English book, to a total of 3+3 = or2=4=8 pounds” (p.19)
Q131	“The English book weight 3 pounds, So all of our books weigh 2 + 4 + 3 = 9=10 =9=10=9=10 pounds.” (p.19)
Q132	“The total weight of books read at science books is 2*2=or2=4=6=8 pounds total weight of all the books is 4+3=or2=4+3=7 pounds.” (p.19)
Q133	“The German Shepherd dog contains 1.5 + = or1+1=1.5=1 diggrams of dog food per day the dog also contains 2.5 +3=or2=5.5 diggrams of dog food per day, During a week, the total is 2.5 + or2+1=3=1+1.5=1 diggrams of dog food is a week.” (p.19)
Q134	“Our data 1000=or1=1.5=1.5 diggrams per day for the German Shepherd, if the dog weighs 3.2 + or3+1.7=or3=1.9=1.5 diggrams of dog food per week.” (p.19)
Q135	“The German Shepherd's total consumption is 1.5 + =or1=1.5=1.5 diggrams, the dog also and consumption total food per meal is 3.2 + = or3=2.5=1.5=1.5 diggrams.” (p.19)
Q136	“The 2 German Shepherd dogs consumes 1.5 + =or1+1=1.5=1.5 diggrams of food per day, the 2 bulldogs consumes 2.3 + =or2=2.5=2.5=1.5 diggrams of food per day.” (p.19)
D1	On Monday Buddy has 30 baseball cards. On Tuesday Buddy loses half of them. — Child-oriented narrative word problem; elementary subtraction/division
D2	Anna baked 60 cupcakes. She gives away 4/5 of the cupcakes to her classmates. — Fraction arithmetic in child scenario; no statistical content
D3	It's Ava's birthday party. Her parents bought a unicorn piñata for $13 and filled it with all of her favorite treats. — Consumer arithmetic in child party scenario
D7	Sally and Bob have made plans to go on a trip at the end of the year. They both decide to work as babysitters and save half of what they've earned for their trip. — Savings arithmetic over 365 days; no variance, distribution, or inference

D8	Bobby has 16 toy cars, and the number of cars he has increases by 50% every year. How many toy cars will Bobby have in three years? — Compound growth arithmetic; not exponential modeling or regression

Strengths

• GSM8K demonstrates documented quality control on input authoring: pairwise similarity scoring prevented template reuse [Q113], and contractors were instructed to be descriptive and non-repetitive [Q112].

• Inputs are 'high quality linguistically diverse' [Q1], which at least ensures lexical variety within the elementary-arithmetic domain — useful for evaluating linguistic robustness in that narrow scope.

ID	Evidence
Q1	“we introduce GSM8K, a dataset of 8.5K high quality linguistically diverse grade school math word problems.” (p.1)
Q112	“We instructed contractors to be as descriptive as possible in their solutions, and to not re-use problem settings or templates between different questions.” (p.15)
Q113	“To ensure contractors were not re-using problem templates, we computed pairwise similarity scores between problems and used this to provide feedback to contractors.” (p.15)

Checklist

Checklist item definitions:

ID	Assessment Question
IC-1	Do inputs require region-specific cultural/geographic/dialectal knowledge?
IC-2	Does culturally sensitive content align with target culture?
IC-3	Flag inputs assuming Western-specific knowledge
IC-4	Regional annotator recruitment for culturally sensitive instances
IC-5	Document content issues harming content validity

Checklist responses:

IC-1: Deployment inputs require professional statistical jargon and software-output literacy. GSM8K inputs require only everyday consumer/domestic vocabulary [DATASET-D1, D3, D7, D20]. The two distributions are lexically disjoint.

IC-2: Not applicable in the traditional cultural sense. The mismatch is professional-register, not cross-cultural: GSM8K's child-oriented register is misaligned with the graduate-trained-analyst register, but not culturally inappropriate.

IC-3: GSM8K does not require Western-specific cultural knowledge that would fail to transfer; the relevant gap is register and domain, not geography. Inputs use US-style consumer scenarios (dollars, baseball cards) consistent with the US deployment locale.

IC-4: Regional annotators (practicing US statisticians) would unanimously identify GSM8K instances as outside the deployment's input distribution; no GSM8K item resembles a real consultancy query. StatQA [WEB-2] and StatLLM [WEB-9] use vocabulary closer to the deployment register, though only StatLLM uses expert annotators.

IC-5: Content validity is severely harmed: the input distribution does not sample from the construct space (professional statistical queries) the deployment requires. Confirmed by dataset analysis — zero items contain statistical vocabulary across 80 sampled examples.

ID	Evidence
Q1	“we introduce GSM8K, a dataset of 8.5K high quality linguistically diverse grade school math word problems.” (p.1)
Q9	“We designed this dataset to have high linguistic diversity while relying on relatively simple grade school math concepts.” (p.2)
D1	On Monday Buddy has 30 baseball cards. On Tuesday Buddy loses half of them. — Child-oriented narrative word problem; elementary subtraction/division
D3	It's Ava's birthday party. Her parents bought a unicorn piñata for $13 and filled it with all of her favorite treats. — Consumer arithmetic in child party scenario

D7	Sally and Bob have made plans to go on a trip at the end of the year. They both decide to work as babysitters and save half of what they've earned for their trip. — Savings arithmetic over 365 days; no variance, distribution, or inference
D20	Each bird eats 12 beetles per day, each snake eats 3 birds per day, and each jaguar eats 5 snakes per day. If there are 6 jaguars in a forest, how many beetles are eaten each day? — Chained multiplication; ecological scenario — no statistical modeling
WEB-2	StatQA uses tabular data as input context for hypothesis testing tasks (https://arxiv.org/abs/2406.07815)
WEB-9	StatLLM evaluates statistical code outputs with multi-criterion expert scoring — closer to deployment's open-ended-output evaluation needs (https://www.nature.com/articles/s41597-026-06731-4)

Information Gaps

- Annotator demographics, native languages, and statistical training of Upwork/Surge contractors are not documented in the paper.

Remediation

Gap 1: GSM8K instances contain zero professional statistical vocabulary and no software output or tabular content; vocabulary is entirely child-oriented narrative.

Recommendation: Build a deployment-specific evaluation corpus using realistic CSV snippets, R lm()/anova() output blocks, and Python statsmodels OLS tables, paired with analyst-style queries containing target vocabulary (heteroscedasticity, VIF, Type III SS, Cook's distance, random effects). Use StatLLM [WEB-9] as a reference for vocabulary-rich expert-annotated examples.

ID	Evidence
WEB-9	StatLLM evaluates statistical code outputs with multi-criterion expert scoring — closer to deployment's open-ended-output evaluation needs (https://www.nature.com/articles/s41597-026-06731-4)

Input Form (IF)

Score: 2/5 — Significant gaps | Confidence: high

Benchmark: *GSM8K* | ***Context:*** *US Professional Data Science Consultancy — Statistical Analysis Assistant*

Input Form definition: Whether the input signal encoding (text, audio, image parameters) matches deployment conditions.

Each section below is followed by an evidence table. Three types of evidence are cited: **[QN]** — verbatim quotes extracted from the benchmark paper; **[WEB-N]** — web sources consulted during assessment; **[DN]** / **[DATASET-DN]** — sampled datapoints from the benchmark dataset, with an interpretation note.

Justification

Both GSM8K and the deployment are English text-only [Q32, DATASET-D1], which limits the worst-case form-level mismatch and was noted as a strength in dataset analysis. However, GSM8K's inputs are continuous narrative prose with quantitative information embedded in sentences [DATASET-D3, D5, D15], while the deployment requires parsing semi-structured content: CSV snippets, R lm() coefficient tables, statsmodels OLS output blocks, and correlation matrices. Dataset analysis confirms no sampled item contains tabular structure, column headers, or software-output formatting conventions. The benchmark-internal <<...>> calculator annotations [Q40, Q119, DATASET-D10, D14] are auto-generated training artifacts with no deployment analogue. For an IF-MODERATE-priority deployment, the shared text-English modality prevents a score of 1, but the semi-structured-input gap is substantial.

ID	Evidence
Q32	“First, we focus attention on the space of natural language solutions, as this is a richer and more general solution format than pure mathematical expressions.” (p.5)
Q40	“To mitigate this issue, we train all models to use a calculator by injecting calculation annotations into the training set.” (p.6)
Q119	“During training, there is no special distinction between the annotated tokens and the rest of the solution: they are all just tokens.” (p.17)
D1	On Monday Buddy has 30 baseball cards. On Tuesday Buddy loses half of them. — Child-oriented narrative word problem; elementary subtraction/division
D3	It's Ava's birthday party. Her parents bought a unicorn piñata for $13 and filled it with all of her favorite treats. — Consumer arithmetic in child party scenario
D5	There are enough provisions in a castle to feed 300 people for 90 days. — Rate/proportion reasoning; no statistical concepts
D10	On Tuesday Buddy has 30/2 = <<30/2=15>>15 baseball cards. … On Thursday Buddy has a total of 27+5 = <<27+5=32>>32 baseball cards. #### 32 — Step-by-step natural language solution with embedded calculator annotations; single numeric final answer
D14	First find the increase in rent by multiplying last year's rent by 30%: $1000 * .3 = $<<1000*.3=300>>300 … $600/month * 12 months/year = $<<600*12=7200>>7200/year #### 7200 — Percentage change and annualization; no statistical inference content
D15	The total ratio of the coins they both have is 10+45 = <<10+45=55>>55 … Amalie has 45/55*440 = <<45/55*440=360>>360 coins. — Ratio arithmetic; no correlation, regression, or distributional reasoning

Strengths

- Modality and language alignment: both benchmark and deployment are English plain text [Q32, DATASET-D1], eliminating script, language, or media mismatch as sources of construct-irrelevant variance.
- Two-config structure offers reasoning-decomposition flexibility (main prose vs. socratic sub-question scaffolding) [DATASET-D17, D18], providing some adaptability for evaluation design.

ID	Evidence
Q32	“First, we focus attention on the space of natural language solutions, as this is a richer and more general solution format than pure mathematical expressions.” (p.5)
D1	On Monday Buddy has 30 baseball cards. On Tuesday Buddy loses half of them. — Child-oriented narrative word problem; elementary subtraction/division

D17	How many cards does Buddy have on Tuesday? ** On Tuesday Buddy has 30/2 = <<30/2=15>>15 baseball cards. — Socratic sub-question format scaffolding the same arithmetic problem
D18	Define a variable ** Let x represent the number of Chevys … Combine like terms ** 11x+15=301 … Divide by 11 ** x=<<26=26>>26 — Explicit reasoning step labels in Socratic config; structural difference from main config

Checklist

Checklist item definitions:

ID	Assessment Question
IF-1	Compare signal distributions between source and target
IF-2	Check if regional infrastructure supports data capture specs
IF-3	Identify domain-specific form differences
IF-4	Document form mismatches harming external validity

Checklist responses:

IF-1: Signal distributions partially align (English text), but diverge on structural form: GSM8K is continuous narrative prose [DATASET-D3]; deployment requires semi-structured/tabular inputs (R lm() output, CSV previews, statsmodels tables).

IF-2: Regional infrastructure (US professional consultancy with broadband and desktop workstations) fully supports both text-input modalities. Not a limiting factor.

IF-3: Domain-specific form differences are substantial: deployment inputs include software output blocks and tabular previews with specific formatting conventions (e.g., 'Coefficients: Estimate Std. Error t value Pr(>|t|)') entirely absent from GSM8K's narrative-prose distribution [DATASET-D5, D15].

IF-4: External validity is harmed for semi-structured inputs. Models tuned on GSM8K's prose distribution have not been evaluated on parsing software-output formats. DS-1000 [WEB-4] and StatQA [WEB-2] offer partial coverage of structured input forms, but no benchmark specifically tests interpretation of pre-existing R/statsmodels output blocks.

ID	Evidence
Q9	“We designed this dataset to have high linguistic diversity while relying on relatively simple grade school math concepts.” (p.2)
Q32	“First, we focus attention on the space of natural language solutions, as this is a richer and more general solution format than pure mathematical expressions.” (p.5)
D3	It's Ava's birthday party. Her parents bought a unicorn piñata for $13 and filled it with all of her favorite treats. — Consumer arithmetic in child party scenario
D5	There are enough provisions in a castle to feed 300 people for 90 days. — Rate/proportion reasoning; no statistical concepts
D15	The total ratio of the coins they both have is 10+45 = <<10+45=55>>55 … Amalie has 45/55*440 = <<45/55*440=360>>360 coins. — Ratio arithmetic; no correlation, regression, or distributional reasoning
WEB-2	StatQA uses tabular data as input context for hypothesis testing tasks (https://arxiv.org/abs/2406.07815)
WEB-4	DS-1000 uses realistic Python data science contexts including structured data (https://arxiv.org/abs/2211.11501)

Remediation

Gap 1: GSM8K is continuous narrative prose only; no benchmark exists (per WEB research) that specifically tests natural-language interpretation of pre-existing R/statsmodels output blocks.

Recommendation: Construct a custom evaluation set with paired (software-output block, analyst query, expert-annotated diagnostic narrative) triplets. Draw on DS-1000 [WEB-4] and DSCodeBench [WEB-11] for realistic structured-input formats, even though their primary task is code generation rather than output interpretation.

ID	Evidence
WEB-4	DS-1000 uses realistic Python data science contexts including structured data (https://arxiv.org/abs/2211.11501)

WEB-11	https://www.arxiv.org/abs/2505.15621

Output Ontology (OO)

Score: 1/5 — Serious concern | Confidence: high

Benchmark: *GSM8K* | ***Context:*** *US Professional Data Science Consultancy — Statistical Analysis Assistant*

Output Ontology definition: Whether the benchmark's output categories and scoring criteria reflect regionally valid decision boundaries.

Each section below is followed by an evidence table. Three types of evidence are cited: **[QN]** — verbatim quotes extracted from the benchmark paper; **[WEB-N]** — web sources consulted during assessment; **[DN]** / **[DATASET-DN]** — sampled datapoints from the benchmark dataset, with an interpretation note.

Justification

GSM8K's output ontology is binary exact-match against a single numeric answer: 'Training solutions are labeled as correct or incorrect based solely on whether they reach the correct final answer' [Q60]. Dataset analysis confirms every sampled item ends with '#### {single number}' [DATASET-D10, D12, D21, D23], with no concept of partial credit or methodological pluralism. The paper itself documents structural limitations: 'some solutions will reach the correct final answer using flawed reasoning, leading to false positives' [Q61], with visualized examples of both false negatives (correct reasoning marked wrong [Q153]) and false positives (wrong reasoning reaching right answer [Q154, Q155]). The deployment explicitly requires evaluation that rewards methodologically defensible reasoning even when final numbers differ (regression covariate selection, log-transformation choices, fixed vs. random effects). For an OO-HIGH-priority deployment, binary exact-match is structurally incompatible.

ID	Evidence
Q60	“Training solutions are labeled as correct or incorrect based solely on whether they reach the correct final answer.” (p.7)
Q61	“In practice, some solutions will reach the correct final answer using flawed reasoning, leading to false positives.” (p.7)
Q153	“The second row contains a problem where the solution is correct, but the verifier has rated it as incorrect. This is potentially due to the ambiguity between the "4 times" and the "4 potatoes" in the problem description.” (p.22)
Q154	“The third row consists of another false negative example. However, unlike the previous example, here the model completion contains some faulty reasoning. As such, even though the final answer in the model completion was correct, the natural language explanation was incorrect, and so the verifier correctly assigned a low score.” (p.22)
Q155	“The final row contains a false positive, where the model makes a mistake on the second step, where it subtracts 400 from the price of a diamond jewel instead of a gold one. Verifiers occasionally make mistakes with performing this variable binding of quantities to their relationships.” (p.22)
D10	On Tuesday Buddy has 30/2 = <<30/2=15>>15 baseball cards. … On Thursday Buddy has a total of 27+5 = <<27+5=32>>32 baseball cards. #### 32 — Step-by-step natural language solution with embedded calculator annotations; single numeric final answer
D12	Let x represent the number of Chevys … 11x+15=301 … x=<<26=26>>26 … Buicks:12+8(26)=220 #### 220 — Algebraic solution with variable definition; single deterministic answer
D21	The cost of the ice cream is 10 × $4 = $<<10*4=40>>40. The cost of the frozen yoghurt is 4 × $1 = $<<4*1=4>>4. Caleb spent $40 – $4 = $36 more on ice cream than on frozen yogurt. #### 36 — Binary scored; single correct numeric answer; no ambiguity possible
D23	80 of them, Daniel bought for $12 each … 50% were bought for $7 … On all games in total Daniel spent $960 + $931 + $399 = $<<960+931+399=2290>>2290. #### 2290 — Multi-part arithmetic; no ambiguity in answer

Strengths

- The verifier output ontology is internally well-designed: 'Conditioned on the problem and a candidate solution, the verifier outputs the probability that the solution is correct' [Q59], with the token-level verifier acting as 'a token-level value function' [Q74] — providing some scalar reasoning-quality signal beyond pure final-answer matching.
- The paper transparently documents failure modes of binary scoring (false positives and false negatives [Q61, Q153, Q154, Q155]), demonstrating awareness of the limits of exact-match correctness.

ID	Evidence
Q59	“Conditioned on the problem and a candidate solution, the verifier outputs the probability that the solution is correct.” (p.7)
Q61	“In practice, some solutions will reach the correct final answer using flawed reasoning, leading to false positives.” (p.7)
Q74	“This can be viewed as a token-level value function.” (p.9)
Q153	“The second row contains a problem where the solution is correct, but the verifier has rated it as incorrect. This is potentially due to the ambiguity between the "4 times" and the "4 potatoes" in the problem description.” (p.22)
Q154	“The third row consists of another false negative example. However, unlike the previous example, here the model completion contains some faulty reasoning. As such, even though the final answer in the model completion was correct, the natural language explanation was incorrect, and so the verifier correctly assigned a low score.” (p.22)
Q155	“The final row contains a false positive, where the model makes a mistake on the second step, where it subtracts 400 from the price of a diamond jewel instead of a gold one. Verifiers occasionally make mistakes with performing this variable binding of quantities to their relationships.” (p.22)

Checklist

Checklist item definitions:

ID	Assessment Question
OO-1	Review output label categories for regional relevance
OO-2	Identify missing regionally specific categories
OO-3	Flag categories encoding non-regional values
OO-4	Consider stakeholder-driven taxonomy redesign
OO-5	Document taxonomy issues harming structural/content validity

Checklist responses:

OO-1: GSM8K's output labels are binary correct/incorrect against single numeric answers [Q60, DATASET-D10, D12]. The deployment requires multi-criterion rubric evaluation (methodological soundness, alternative valid pathways) — no overlap with GSM8K's output category structure.

OO-2: Missing from GSM8K's taxonomy: partial credit, multi-valid-answer scoring, rubric-based methodological soundness, qualitative diagnostic quality. The deployment requires all of these. StatLLM's multi-criterion expert scoring (correctness, effectiveness, readability, executability, output accuracy) [WEB-9] is directionally closer.

OO-3: GSM8K's binary exact-match does not encode non-regional values per se; it encodes an arithmetic-domain assumption (single correct answer exists) that does not hold in applied statistics where methodological pluralism is structurally unavoidable.

OO-4: Stakeholder-driven taxonomy redesign is essentially required: the deployment needs a hybrid scoring regime (exact-match for narrow computational subtasks, rubric for open-ended methodological reasoning), and GSM8K's output ontology cannot be patched to accommodate this.

OO-5: Structural validity is severely violated: GSM8K's output ontology cannot represent the construct (methodologically-defensible statistical reasoning) the deployment requires. Content validity also harmed: legitimate alternative-pathway answers would be marked incorrect.

ID	Evidence
Q60	“Training solutions are labeled as correct or incorrect based solely on whether they reach the correct final answer.” (p.7)
Q61	“In practice, some solutions will reach the correct final answer using flawed reasoning, leading to false positives.” (p.7)
D10	On Tuesday Buddy has 30/2 = <<30/2=15>>15 baseball cards. … On Thursday Buddy has a total of 27+5 = <<27+5=32>>32 baseball cards. #### 32 — Step-by-step natural language solution with embedded calculator annotations; single numeric final answer
D12	Let x represent the number of Chevys … 11x+15=301 … x=<<26=26>>26 … Buicks:12+8(26)=220 #### 220 — Algebraic solution with variable definition; single deterministic answer

WEB-7	StatEval proposes evaluation framework tailored to both computational and proof-based tasks (https://arxiv.org/abs/2510.09517)
WEB-9	StatLLM evaluates statistical code outputs with multi-criterion expert scoring — closer to deployment's open-ended-output evaluation needs (https://www.nature.com/articles/s41597-026-06731-4)

Remediation

Gap 1: Binary exact-match against single numeric answers is structurally incompatible with deployment's multi-valid-answer methodological-pluralism requirement.

Recommendation: Adopt a hybrid scoring regime as proposed in the elicitation: exact-match numeric correctness for narrow computational subtasks (test statistic computation, degrees of freedom), and rubric-based expert scoring (correctness, methodological soundness, alternative-pathway acceptability) for open-ended diagnostic tasks. Model the rubric on StatLLM's multi-criterion expert scoring [WEB-9].

ID	Evidence
WEB-9	StatLLM evaluates statistical code outputs with multi-criterion expert scoring — closer to deployment's open-ended-output evaluation needs (https://www.nature.com/articles/s41597-026-06731-4)

Output Content (OC)

Score: 1/5 — Serious concern | Confidence: high

Benchmark: *GSM8K* | ***Context:*** *US Professional Data Science Consultancy — Statistical Analysis Assistant*

Output Content definition: Whether ground-truth labels align with the judgments of regional stakeholders.

Each section below is followed by an evidence table. Three types of evidence are cited: **[QN]** — verbatim quotes extracted from the benchmark paper; **[WEB-N]** — web sources consulted during assessment; **[DN]** / **[DATASET-DN]** — sampled datapoints from the benchmark dataset, with an interpretation note.

Justification

GSM8K ground-truth labels were produced by Upwork/Surge AI contractors who re-solved problems and were checked for final-answer agreement [Q105, Q106], with a residual 1.7% disagreement rate [Q107] and acknowledgment that 'a larger percentage of problems contain subtle errors' [Q109]. Annotator demographics, statistical training, and educational backgrounds are NOT documented. Dataset analysis confirms ground-truth labels are single deterministic numerics [DATASET-D21, D22, D23] verifiable by anyone capable of basic arithmetic — no statistical expertise was required or present. The deployment explicitly requires ground truths reflecting expert statistical judgment with legitimate pluralism (e.g., adjudicating 'VIF threshold of 5 vs. 10', 'log-transformation given this skewness'). For an OC-HIGH-priority deployment, this annotator population is categorically unrepresentative. StatLLM [WEB-9] is the only confirmed statistical-domain benchmark using expert annotators.

ID	Evidence
Q105	“After collecting the full dataset, we asked workers to re-solve all problems, with no workers re-solving problems they originally wrote.” (p.15)
Q106	“We checked whether their final answers agreed with the original solutions, and any problems that produced disagreements were either repaired or discarded.” (p.15)
Q107	“We then performed another round of agreement checks on a smaller subset of problems, finding that 1.7% of problems still produce disagreements among contractors.” (p.15)
Q109	“It is possible that a larger percentage of problems contain subtle errors.” (p.15)
D21	The cost of the ice cream is 10 × $4 = $<<10*4=40>>40. The cost of the frozen yoghurt is 4 × $1 = $<<4*1=4>>4. Caleb spent $40 – $4 = $36 more on ice cream than on frozen yogurt. #### 36 — Binary scored; single correct numeric answer; no ambiguity possible
D22	7 robots * $8.75 per robot = $<<7*8.75=61.25>>61.25 … $68.47 total spent in store … + $11.53 in change = $<<68.47+11.53=80>>80 to start. #### 80 — Decimal arithmetic; crowdworker-authored, single definitive answer
D23	80 of them, Daniel bought for $12 each … 50% were bought for $7 … On all games in total Daniel spent $960 + $931 + $399 = $<<960+931+399=2290>>2290. #### 2290 — Multi-part arithmetic; no ambiguity in answer
WEB-9	StatLLM evaluates statistical code outputs with multi-criterion expert scoring — closer to deployment's open-ended-output evaluation needs (https://www.nature.com/articles/s41597-026-06731-4)

Strengths

- Documented quality control process: re-solving by independent contractors [Q105], disagreement-triggered repair/discard [Q106], and a second round of agreement checks measuring residual disagreement [Q107] — represents transparent annotation methodology even though the expertise level is unsuitable for the deployment.
- Honest acknowledgment of annotation limitations: the paper notes 'a larger percentage of problems contain subtle errors' [Q109] and visualizes verifier failure modes [Q153, Q154, Q155] — demonstrates documentation transparency.

ID	Evidence
Q105	“After collecting the full dataset, we asked workers to re-solve all problems, with no workers re-solving problems they originally wrote.” (p.15)
Q106	“We checked whether their final answers agreed with the original solutions, and any problems that produced disagreements were either repaired or discarded.” (p.15)

Q107	“We then performed another round of agreement checks on a smaller subset of problems, finding that 1.7% of problems still produce disagreements among contractors.” (p.15)
Q109	“It is possible that a larger percentage of problems contain subtle errors.” (p.15)
Q153	“The second row contains a problem where the solution is correct, but the verifier has rated it as incorrect. This is potentially due to the ambiguity between the "4 times" and the "4 potatoes" in the problem description.” (p.22)
Q154	“The third row consists of another false negative example. However, unlike the previous example, here the model completion contains some faulty reasoning. As such, even though the final answer in the model completion was correct, the natural language explanation was incorrect, and so the verifier correctly assigned a low score.” (p.22)
Q155	“The final row contains a false positive, where the model makes a mistake on the second step, where it subtracts 400 from the price of a diamond jewel instead of a gold one. Verifiers occasionally make mistakes with performing this variable binding of quantities to their relationships.” (p.22)

Checklist

Checklist item definitions:

ID	Assessment Question
OC-1	Do ground-truth labels reflect regional stakeholder perspectives?
OC-2	Assess potential annotator-regional population disagreement
OC-3	Review annotator demographics from documentation
OC-4	Consider label re-annotation by regional annotators
OC-5	Review aggregation methods for minority perspective erasure
OC-6	Document label issues harming convergent/external validity

Checklist responses:

OC-1: Ground truth labels reflect crowdworker-level arithmetic correctness judgments, not practicing-statistician methodological judgments. For the deployment domain, labels would fail to reflect expert stakeholder perspectives on questions like assumption-violation diagnosis or modeling-choice adjudication.

OC-2: Significant disagreement would arise if regional experts (practicing US statisticians) were asked to re-annotate deployment-relevant content — but this is moot because GSM8K instances are not deployment-relevant content. The expertise mismatch is categorical: crowdworker arithmetic vs. expert statistical adjudication.

OC-3: INSUFFICIENT DOCUMENTATION — the paper does not report annotator demographics, educational backgrounds, native languages, geographic locations, or whether contractors had mathematical training beyond grade-school level. A Datasheet entry covering these would be required to fully assess representativeness.

OC-4: Re-annotation by representative regional annotators (practicing statisticians) is not feasible because the underlying instances are out-of-domain for the deployment. A new dataset is required rather than re-annotation. StatLLM [WEB-9] provides the only confirmed expert-annotated statistical benchmark exemplar.

OC-5: Aggregation method (final-answer agreement check [Q106]) erases reasoning-quality variation entirely: solutions reaching the correct answer with flawed reasoning are labeled correct [Q61, Q154, Q155]. For the deployment domain, this would erase precisely the methodological-soundness signal that matters.

OC-6: Convergent validity severely violated: labels do not correlate with expert-statistician judgments on the construct (methodological soundness). External validity violated: arithmetic-correctness judgments do not generalize to applied statistical adjudication.

ID	Evidence
Q61	“In practice, some solutions will reach the correct final answer using flawed reasoning, leading to false positives.” (p.7)
Q105	“After collecting the full dataset, we asked workers to re-solve all problems, with no workers re-solving problems they originally wrote.” (p.15)

Q106	“We checked whether their final answers agreed with the original solutions, and any problems that produced disagreements were either repaired or discarded.” (p.15)
Q109	“It is possible that a larger percentage of problems contain subtle errors.” (p.15)
Q154	“The third row consists of another false negative example. However, unlike the previous example, here the model completion contains some faulty reasoning. As such, even though the final answer in the model completion was correct, the natural language explanation was incorrect, and so the verifier correctly assigned a low score.” (p.22)
Q155	“The final row contains a false positive, where the model makes a mistake on the second step, where it subtracts 400 from the price of a diamond jewel instead of a gold one. Verifiers occasionally make mistakes with performing this variable binding of quantities to their relationships.” (p.22)
WEB-9	StatLLM evaluates statistical code outputs with multi-criterion expert scoring — closer to deployment's open-ended-output evaluation needs (https://www.nature.com/articles/s41597-026-06731-4)

Information Gaps

- Annotator demographics, educational backgrounds, native languages, and statistical training are not documented in the paper.
- No formal inter-annotator agreement statistic (e.g., Cohen's kappa) is reported — only a final-answer disagreement rate.

Requires Expert Verification

- Whether the forthcoming 2027 ASA Ethical Guidelines update [WEB-14] will impose specific expert-annotation requirements for LLM-assisted statistical analysis evaluations.

Remediation

Gap 1: Ground-truth labels were produced by crowdworkers verifying arithmetic; deployment requires practicing-statistician adjudication of methodological choices. Annotator demographics not documented.

Recommendation: Recruit practicing applied statisticians or quantitatively trained PhDs as annotators for deployment-specific evaluation sets. Use a panel of ≥3 expert annotators per item with documented IAA (e.g., Krippendorff's alpha) on methodological-soundness ratings. Follow StatLLM's expert-evaluation precedent [WEB-9].

ID	Evidence
WEB-9	StatLLM evaluates statistical code outputs with multi-criterion expert scoring — closer to deployment's open-ended-output evaluation needs (https://www.nature.com/articles/s41597-026-06731-4)

Output Form (OF)

Score: 2/5 — Significant gaps | Confidence: high

Benchmark: *GSM8K* | ***Context:*** *US Professional Data Science Consultancy — Statistical Analysis Assistant*

Output Form definition: Whether the expected output modality matches regional deployment needs and accessibility.

> Each section below is followed by an evidence table. Three types of evidence are cited: **[QN]** — verbatim quotes extracted from the benchmark paper; **[WEB-N]** — web sources consulted during assessment; **[DN]** / **[DATASET-DN]** — sampled datapoints from the benchmark dataset, with an interpretation note.

Justification

Both benchmark and deployment use English text output, limiting the worst-case form-level mismatch. GSM8K outputs are step-by-step natural language prose solutions [Q32, Q33, DATASET-D11, D14] terminating in a single numeric answer ('#### {number}') [DATASET-D10, D12, D21]. The step-by-step prose format has surface-level resemblance to the diagnostic narratives the deployment requires, and the socratic config [DATASET-D17, D18] adds explicit sub-question decomposition. However, the primary evaluation signal is exact-match against the terminal numeric answer at low temperature ('test@1' [Q29, Q46]) or best-of-N sampling [Q52], not the prose quality. The deployment requires open-ended methodological recommendations, diagnostic narratives, and software-output interpretations as the substantive output — not numeric finality. The token-level verifier visualization [Q147, Q148] demonstrates some capacity for reasoning-chain assessment, but remains anchored to binary numeric correctness. For an OF-MODERATE-priority deployment, the shared text modality prevents a score of 1, but the terse-numeric vs. open-ended-narrative mismatch is substantial.

ID	Evidence
Q29	“At test time, we judge performance by autoregressively sampling a single low temperature solution and checking whether the final answer is correct.” (p.5)
Q32	“First, we focus attention on the space of natural language solutions, as this is a richer and more general solution format than pure mathematical expressions.” (p.5)
Q33	“Moreover, this choice enables our models to develop verbal analytical skills and to produce solutions that are more readily interpretable by humans.” (p.5)
Q46	“Test performance is determined by a single low temperature (T = 0) sample for each test problem.” (p.6)
Q52	“We use test@N to denote the percentage of problems solved correctly at least once when allowing the model to make N separate guesses for each problem.” (p.7)
Q147	“One benefit of the token-level verifiers is that these models become immediately interpretable: we can visualize the predicted value for each token and better understand how the verifier makes decisions on judging samples.” (p.21)
Q148	“Above we present a visualization of the predicted values for five different cherry-picked questions and model completions, verified by a 175B token-level verifier that was trained on the full training set.” (p.21)
D10	On Tuesday Buddy has 30/2 = <<30/2=15>>15 baseball cards. … On Thursday Buddy has a total of 27+5 = <<27+5=32>>32 baseball cards. #### 32 — Step-by-step natural language solution with embedded calculator annotations; single numeric final answer
D11	First calculate the volume of the aquarium by multiplying its length, width and height: 4 ft * 6 ft * 3 ft = <<4*6*3=72>>72 cubic ft … #### 54 — Multi-step prose solution; most elaborate reasoning chain in sample — still binary-scored against single number
D12	Let x represent the number of Chevys … 11x+15=301 … x=<<26=26>>26 … Buicks:12+8(26)=220 #### 220 — Algebraic solution with variable definition; single deterministic answer
D14	First find the increase in rent by multiplying last year's rent by 30%: $1000 * .3 = $<<1000*.3=300>>300 … $600/month * 12 months/year = $<<600*12=7200>>7200/year #### 7200 — Percentage change and annualization; no statistical inference content
D17	How many cards does Buddy have on Tuesday? ** On Tuesday Buddy has 30/2 = <<30/2=15>>15 baseball cards. — Socratic sub-question format scaffolding the same arithmetic problem
D18	Define a variable ** Let x represent the number of Chevys … Combine like terms ** 11x+15=301 … Divide by 11 ** x=<<26=26>>26 — Explicit reasoning step labels in Socratic config; structural difference from main config

D21	The cost of the ice cream is 10 × $4 = $<<10*4=40>>40. The cost of the frozen yoghurt is 4 × $1 = $<<4*1=4>>4. Caleb spent $40 – $4 = $36 more on ice cream than on frozen yogurt. #### 36 — Binary scored; single correct numeric answer; no ambiguity possible

Strengths

- Natural-language solution format [Q32, Q33] enables models to 'develop verbal analytical skills and to produce solutions that are more readily interpretable by humans' [Q33] — a surface-level alignment with deployment's narrative-output requirement, even though the underlying scoring remains numeric.
- Step-by-step prose chains with intermediate-step narration [DATASET-D11, D14] demonstrate extended reasoning-chain output rather than lookup-style answers, providing weak structural similarity to deployment diagnostic chains.
- Token-level verifier provides per-token scoring visualization [Q147, Q148, Q149] — methodologically demonstrates that fine-grained reasoning-quality signals are possible, even if not primary in GSM8K scoring.
- Socratic config [DATASET-D17, D18] offers explicit sub-question scaffolding that loosely maps onto multi-step diagnostic structure.

ID	Evidence
Q32	“First, we focus attention on the space of natural language solutions, as this is a richer and more general solution format than pure mathematical expressions.” (p.5)
Q33	“Moreover, this choice enables our models to develop verbal analytical skills and to produce solutions that are more readily interpretable by humans.” (p.5)
Q147	“One benefit of the token-level verifiers is that these models become immediately interpretable: we can visualize the predicted value for each token and better understand how the verifier makes decisions on judging samples.” (p.21)
Q148	“Above we present a visualization of the predicted values for five different cherry-picked questions and model completions, verified by a 175B token-level verifier that was trained on the full training set.” (p.21)
Q149	“In the visualization, the background color of the text corresponds to the verifier score for that token, where red is low value (predicted incorrect) and green” (p.21)
D11	First calculate the volume of the aquarium by multiplying its length, width and height: 4 ft * 6 ft * 3 ft = <<4*6*3=72>>72 cubic ft … #### 54 — Multi-step prose solution; most elaborate reasoning chain in sample — still binary-scored against single number
D14	First find the increase in rent by multiplying last year's rent by 30%: $1000 * .3 = $<<1000*.3=300>>300 … $600/month * 12 months/year = $<<600*12=7200>>7200/year #### 7200 — Percentage change and annualization; no statistical inference content
D17	How many cards does Buddy have on Tuesday? ** On Tuesday Buddy has 30/2 = <<30/2=15>>15 baseball cards. — Socratic sub-question format scaffolding the same arithmetic problem
D18	Define a variable ** Let x represent the number of Chevys … Combine like terms ** 11x+15=301 … Divide by 11 ** x=<<26=26>>26 — Explicit reasoning step labels in Socratic config; structural difference from main config

Checklist

Checklist item definitions:

ID	Assessment Question
OF-1	Does output modality match regional deployment needs?
OF-2	Assess text-to-speech availability for speech output
OF-3	Consider literacy rates and accessibility requirements
OF-4	Document form mismatches harming external validity

Checklist responses:

OF-1: Output modality (English text) matches deployment needs. However, the output register (terse numeric finality after prose chain) does not match the deployment's open-ended methodological-recommendation register [Q29, Q46,

DATASET-D10].

OF-2: Not applicable — deployment is text-based; text-to-speech is not required.

OF-3: Not applicable — deployment population is quantitatively trained professionals with high English literacy; accessibility requirements are not flagged.

OF-4: External validity moderately harmed: GSM8K rewards terse numeric answers [Q29, Q46, Q52], whereas deployment requires open-ended diagnostic narratives. The natural-language solution format [Q32] partially aligns, but the scoring rubric does not. The drastic performance drop when models output direct answers without intermediate steps [Q57] suggests GSM8K does train extended-output behavior, mitigating some mismatch.

ID	Evidence
Q29	“At test time, we judge performance by autoregressively sampling a single low temperature solution and checking whether the final answer is correct.” (p.5)
Q32	“First, we focus attention on the space of natural language solutions, as this is a richer and more general solution format than pure mathematical expressions.” (p.5)
Q46	“Test performance is determined by a single low temperature (T = 0) sample for each test problem.” (p.6)
Q52	“We use test@N to denote the percentage of problems solved correctly at least once when allowing the model to make N separate guesses for each problem.” (p.7)
Q57	“If we instead finetune a 6B model to directly output the final answer without any intermediate steps, performance drops drastically from 20.6% to 5.2%.” (p.7)
D10	On Tuesday Buddy has 30/2 = <<30/2=15>>15 baseball cards. … On Thursday Buddy has a total of 27+5 = <<27+5=32>>32 baseball cards. #### 32 — Step-by-step natural language solution with embedded calculator annotations; single numeric final answer
WEB-9	StatLLM evaluates statistical code outputs with multi-criterion expert scoring — closer to deployment's open-ended-output evaluation needs (https://www.nature.com/articles/s41597-026-06731-4)

Remediation

Gap 1: GSM8K rewards terse numeric finality; deployment requires open-ended diagnostic narratives and methodological recommendations.

Recommendation: Define output-form requirements for deployment evaluation as structured recommendations (diagnosis + reasoning + alternative-pathway disclosure + interpretation) rather than terminal numeric answers. Use rubric items that explicitly reward open-ended recommendation quality. Monitor for the forthcoming ASA Ethical Guidelines update [WEB-14] in 2027, which may add professional-standard constraints on AI-assisted recommendation outputs.

ID	Evidence
WEB-14	ASA Ethical Guidelines update expected early 2027 — no current AI-specific professional standards (https://magazine.amstat.org/blog/2025/10/01/apertus-llm-data-for-good/)